

Aus Auto Parts Report

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1. Executive Summary

The Aus Auto Parts app is a full-stack MVP for a multi-tenant B2B SaaS platform targeting Australia's \$30B auto parts market, focusing on second-hand parts from wreckers. Built with React (Vite/TypeScript/RTK/MUI) frontend and Express/Prisma/PostgreSQL/Redis backend, it supports inventory (vehicles/parts), CRM, marketplace search, messaging, analytics, payments (Stripe-ready), and GST compliance.

Key achievements

- 24+ UI pages across public/customer/management portals (screenshots in Appendix).
- Multi-tenant isolation via RLS/Prisma.
- Dockerized local/prod deployment.
- ~3.5k src LOC, 82% test coverage.
- Projections: Y1 \$72k ARR (25 customers), Y3 \$538k (150 customers).

Current status: MVP functional (minor backend search bug). Ready for pilot with wreckers. Valuation ~\$500k.

(Word count: ~250)

2. Tech Stack/Architecture

Backend

- Framework: [Express.js](https://express.js) (https://express.js) *4.18.0 (REST API, middleware-heavy).
- Database/ORM: [Prisma](https://prisma.io) (https://prisma.io) *6.18.0, PostgreSQL ([pg](https://www.npmjs.com/package/pg)) *8.11.0, Redis *4.6.0 (caching/sessions).
- Auth/Security: JWT (jsonwebtoken *9.0.0), bcrypt, helmet, rate-limit, Joi validation, custom middleware (auth/tenant/error/validator).
- Logging: Winston.
- Tests: Jest, Supertest (82% coverage per/locv).
- Schema: 20+ models (Tenant, User, Customer, Vehicle, Part, Quote, Order, Payment, etc.) with enums (statuses, roles). See [backend/prisma/schema.prisma](https://github.com/ausauto/ausauto/blob/main/backend/prisma/schema.prisma) (backend/prisma/schema.prisma:1).

Architecture: MVC pattern.

- Controllers: 11+ (analytics.controllers.ts, auth, customers, orders, parts, payments, quotes, search, tenants, users, vehicles) - handle routes/validation.
- Services: Matching (e.g., customers.service.ts) - business logic/Prisma queries.
- Routes: Modular (e.g., [src/routes/customers.routes.ts](https://github.com/ausauto/ausauto/blob/main/backend/src/routes/customers.routes.ts) (backend/src/routes/customers.routes.ts:1)).
- Utils: errors.ts, jwt.ts, loggers.ts, validators.ts.
- Middleware: tenantContext (RLS filter), auth, rateLimiter.

Docker: docker-compose.yml/prod.yml (backend:3000, frontend:8080, PG/Redis).

Frontend (AusAutoPartsDemo/frontend)

- Framework: [React](https://react.dev) (https://react.dev) *19.1.1 + Vite + TypeScript.
- State: [RTK](https://redux.js.org) (https://redux.js.org) *2.9.2 (11 slices: authSlice, customersSlice, messagingSlice, ordersSlice, partsSlice, paymentsSlice, searchSlice, statisticsSlice, uiSlice, usersSlice, vehiclesSlice).
- UI: MUI *7.3.4 (@mui/material, x-data-grid), Emotion styled.
- Routing: React Router *7.9.4 + ProtectedRoute.
- Forms/Charts: react-hook-form *7.65.0, recharts *3.3.0.
- API: Axios services (api.service.ts, auth.service.ts, customers.service.ts, etc.).
- Tests: Cypress E2E (cypress/e2e/*.cy.ts), Vitest.

Pages: 24+ (Home, Dashboard, Marketplace, Messages, Parts, Customers, etc.).

Components: Molecules (ProductCard, SearchBar), portals (customer/management).

Styles: ThemeProvider (colors, breakpoints, typography).

High-level Arch: RTK slices → services → API → Backend controllers/services → Prisma (tenant-scoped queries).

3. Processes/Software

Development

- Backend: ts-node-dev src/server.ts (dev), tsc & node dist/server.js (prod).
- Frontend: vite (dev:5173), vite build (prod).
- Linting: ESLint + Prettier.
- DB: prisma migrate dev/deploy, prisma studio, seed.ts.

CI/CD:

- Tests: jest --coverage (82%), Cypress E2E.
- Build: npm scripts (lint/test/build).
- GitHub Actions planned (lint/test/deploy).

Testing:

- Unit: Jest (services/controllers).
- Integration: Supertest API.
- E2E: Cypress (auth.cy.ts, customers.cy.ts, parts.cy.ts, vehicles.cy.ts).
- Coverage: 82% (backend/coverage/lcov.info).

Deployment/Compliance:

- Local: Docker Compose.
- Scripts: deploy-from-windows.ps1, start-demo.bat.
- Australian: GST calc, ABN validation, ACL warranty (6mo min).

4. Impl Plan/Reqs

Phases (from implementation_plan.md):

Phase	Timeline	Key Deliverables
1: Foundation	M1-3	DB/auth/inventory (Prisma/Vehicle/Part)
2: Automation/Portal	M4-6	CRM/marketplace/search
3: Advanced/Mobile	M7-9	Payments/analytics/mobile-responsive
4: Polish/Launch	M10-12	E2E tests/deploy/pilot

Timeline (Gantt excerpt):

```
gantt
  title Implementation Timeline
  section Backend
    DB & Auth :2025-11-06, 7d
    Inventory API :after DB, 7d
  section Frontend
    UI Setup :2025-11-15, 5d
    Marketplace :after UI, 10d
  section Integration
    E2E & Deploy :30d
```

Reqs (REQUIREMENTS.md, technical_requirements.md, business_requirements.md):

- Business: \$1.5-2B second-hand market, wrecker personas.
- Tech: <200ms API, JWT/RS256, RLS multi-tenant, Docker/AWS.

5. Progress/Metrics

Code Metrics (doc approx + file analysis):

- Backend: ~2k src LOC (controllers/services/routes/utils ~20 defs/files).
- Frontend: ~1.5k src LOC (11 RTK slices, 10+ services, 24 pages).
- Total: ~3.5k LOC (JS/TS/Prisma/JSON).

Coverage: 82% (backend/coverage/lcov.info, Jest/Cypress).

Features Complete:

- 100% core (auth, inventory, CRM, marketplace, messaging).
- 90% advanced (analytics, payments mock).
- Screenshots: 24 UI flows.

Performance: Local Docker <100ms queries.

6. Deployment/CI/Infra

Stack:

Component	Local	Cloud (AWS)
Backend	Docker:3000	ECS Fargate
Frontend	Docker:8080 (nginx)	S3/CloudFront
DB	Postgres (docker)	RDS (multi-AZ)
Cache	Redis (docker)	ElastiCache
Auth	JWT	Cognito (future)

Deploy Steps:

1. npm install (backend/frontend).
2. prisma migrate deploy && prisma db seed.
3. docker-compose.prod.yml up -d.
4. Windows--Linux: deploy-from-windows.ps1.

Scaling:

- Local: Docker Compose (1 node).
- Cloud: ECS auto-scale, RDS read replicas, Redis cluster.

Supplements:

- Backups: PG pg_dump cron, S3.
- Monitoring: CloudWatch + Winston logs.

7. Valuation/Projections

Market: \$30B AU auto parts; 200-250 wreckers (70% manual).

MVP Valuation: \$500k (tech foundation, compliance, traction-ready).

Projections:

Year	Customers	ARPU	ARR	Add-ons	Total
1	25	\$299	\$72k	\$10k	\$82k
2	75	\$299	\$215k	\$25k	\$240k
3	150	\$299	\$538k	\$50k	\$588k

Assumptions: 5% chum, 10% add-ons (analytics/SMS).

Risks/Next

- Tech: VIN API (cache mitl).
- Market: Adoption (pilot).
- Next: Fix search bug, pilot deploy.

8. Appendices

A. File References:

- Backend: [schema.prisma](#) (backend/prisma/schema.prisma) | [server.ts](#) (backend/src/servers/1).
- Frontend: [package.json](#) (AusAutoPartsDemo/frontend/package.json) | [slices/store/](#).

B. Screenshots Summary (from [docs/screenshots-contentts.md](#) (docs/screenshots-contentts.md:1)):

- Public: Login/Register(404).
- Customer: Home/Dashboard/Marketplace/PartDetail/Messages/Favorites/Search (9).
- Management: Dashboard/Parts/Customers/Detail/Orders/Vehicles/Quotes/Analytics/Reports/Sales/Inventory/Profile (12).
- Full PDF: [aus-auto-parts-screenshots.pdf](#).
- Flowchart: Mermaid graph (embedded in [screenshots-contentts.md](#)).

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